SC4001 Neural Networks and Deep Learning

Group Project, Topic E: Object Hallucination in Vision-Language Models

Does a Small Vision-Language Model Look Away When It Hallucinates?

Image Attention, an Inference-Time Attention Boost and Abstention for Object Hallucination in Qwen2-VL-2B

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1. Introduction

Vision language models (VLMs) answer questions about images and describe them in fluent text, but they regularly mention objects that are not in the image. This *object hallucination* makes them unreliable in settings where a wrong statement is worse than no statement. A common explanation is that the language model inside a VLM relies on its text prior (what usually co-occurs with what it has already written) more than on the image [3, 5]. If that is right, hallucinated words should coincide with the model paying *less attention to the image*, and pushing attention back towards the image should help.

We test this explanation on a small open model, Qwen2-VL-2B-Instruct [6], with one focused question: **when Qwen2-VL-2B mentions an object that is not in the image, is it paying less attention to the image at that moment, and can increasing its attention to image tokens at inference time reduce hallucination without hurting answer quality?** We state three hypotheses:

- **H1 (analysis).** Tokens of hallucinated objects receive a lower share of attention on image tokens than tokens of real objects, and the gap is clearest in particular layers.
- **H2 (intervention).** Multiplying attention on image tokens in selected layers lowers hallucination on POPE [1] and CHAIR [2] compared with the unmodified model.
- **H3 (trade-off).** A strong intervention hurts recall, caption length or fluency, so there is a sweet spot, and the adversarial POPE split benefits least.

Our contributions are: (i) a controlled evaluation of Qwen2-VL-2B on POPE and CHAIR with a strict dev/test separation; (ii) a layer-by-layer analysis of image attention at the moment each object word is written, which controls for the position of the word and compares against the model's own token confidence; (iii) a training-free, post-softmax image-attention boost whose strength and layers are chosen by a rule fixed before looking at test data, compared with PAI's attention step tuned under the same rule; (iv) attention-based abstention compared with a confidence baseline; and (v) an analysis of the cases where the method fails. Most prior attention-based work uses 7B-scale LLaVA-style models [3, 4, 5]; a 2B model with dynamic-resolution image tokens is a different and cheaper regime. Well-supported negative results are reported as such.

2. Review of existing techniques

2.1 Measuring object hallucination

CHAIR [2] scores free-form captions on MS-COCO [7]. Words are mapped to the 80 COCO categories with a synonym list, and a mentioned category is hallucinated if it is neither annotated in the image nor named in its reference captions. CHAIR-s is the share of captions with at least one hallucinated object and CHAIR-i the share of object mentions that are hallucinated. Because a model can lower CHAIR by saying less, it must be read together with object recall and caption length. **POPE** [1] instead asks balanced yes/no questions ("Is there a <object> in the image?") about 500 COCO images. Absent objects are sampled at random, among the most frequent COCO objects (popular), or among objects that frequently co-occur with the present ones (adversarial). POPE is stable and cheap but only probes recognition, not generation; we therefore use both.

2.2 Training-free mitigation

VCD [3] contrasts the output distribution given the real image with the distribution given a noise-distorted image, down-weighting tokens that the model would produce anyway from its language prior. **OPERA** [4] observes that hallucinations often follow "summary" tokens on which attention columns pile up, and adds a penalty for this over-trust pattern to beam search, with a retrospective roll-back. **PAI** [5] ("Paying More Attention to Image") is closest to our work. It names the problem *text inertia*, amplifies the pre-softmax attention logits of image tokens for the current token by alpha times their magnitude in layers chosen with an attention-sink criterion, and additionally subtracts the logits of an image-free forward pass; it was evaluated on LLaVA-1.5, MiniGPT-4 and Shikra.

2.3 How our work differs

Our intervention is a *post-softmax* rescaling: in a contiguous band of decoder layers, every attention row whose query comes after the image has its weights on image tokens multiplied by (1 + alpha) and is renormalised. It uses no image-free pass and no contrastive logits, so any effect can be attributed to attention alone, and alpha = 0 reproduces the original model exactly. We apply it to Qwen2-VL-2B, whose number of image tokens depends on the image (here 64 to 256) and whose layers are chosen on a dev set from our own attention analysis. To compare with PAI, we also re-implement its attention step without the image-free pass and tune it with the same dev protocol (Section 3.4). Finally, rather than only intervening, we ask whether the same attention signal can tell the model *when to abstain*, and compare it with the simplest alternative, the model's own answer confidence.

3. Methods

3.1 Model and inference

We use Qwen2-VL-2B-Instruct [6] through Hugging Face Transformers 5.22.2 with eager attention (needed to read attention weights). Qwen2-VL encodes an image with a ViT into a variable number of tokens (one token per 28 x 28 pixel area); we fix the image budget to 64-256 tokens (at most 200,704 pixels) so every method sees identical inputs. Decoding is greedy, so outputs are deterministic. Our sanity checks (E0) showed that **float16 is unusable** for this model in this library version: the residual stream overflows and every attention map and logit from decoder layer 1 onward is NaN, so every answer was "!!!!!". We therefore run in bfloat16, which passed every check (answers correct on 10/10 hand-read questions, attention rows summing to 1 within 4e-3, deterministic outputs). All experiments ran on one NVIDIA RTX 4060 Laptop GPU (8 GB). To make the runs affordable we generate in batches (12 questions or 8 captions, left-padded). Every method is run with identical batches, so comparisons stay controlled.

3.2 Benchmarks, metrics and data protocol

POPE. The three COCO splits (3,000 questions each, 50% "yes") with the suffix " Answer with yes or no.". The first word of the answer is used if it is yes/no; otherwise the original POPE rule applies ("no" if the answer contains "no" or "not"). We report accuracy, precision, recall, F1 and yes-ratio. **CHAIR.** One caption per image with the prompt "Describe this image in detail." (at most 256 new tokens) for 500 COCO val2014 images. We re-implemented the original scoring with the authors' synonym list and two-word rules; ground truth is the union of instance annotations and objects named in the five reference captions. We report CHAIR-s, CHAIR-i, object recall (distinct ground-truth categories mentioned / distinct ground-truth categories) and length.

Dev/test separation. We drew, with a fixed seed, 200 further COCO val2014 images that appear in neither test set. They provide 902 POPE-style questions built with POPE's random-negative protocol, and captions for CHAIR. *Every* choice (the layer band, alpha, the abstention band and its threshold) is made on dev. Each test set was then run once with the frozen settings; later reruns on other GPUs (Appendix A) only checked reproducibility and were not used to choose anything.

3.3 Image-attention share at an object mention (E3)

For layer l, let $a^{(l)}_h(q, k)$ be the attention of head h from query position q to key k. The image-attention share is s_l(q) = mean_h sum_{k in image} $a^{(l)}$_h(q, k), the fraction of attention that q puts on image tokens. For every object mention found by CHAIR in a caption, we re-run the model with teacher forcing on its own caption and record s_l at the position that *predicts* the object's first token (the moment the model decides to write the object) for all 28 layers, together with the log-probability of that token and its position in the caption. Image-token positions are found from the processor output and asserted (count matches the image grid, contiguous, wrapped by the vision start/end tokens). We measure how well low attention identifies hallucinated mentions with the AUROC, with 95% confidence intervals from a bootstrap that resamples whole images. Because later tokens attend less to the image and hallucinations tend to come late in long captions, position is a confounder. We control for it in two ways: AUROC within caption-position quintiles, and cross-validated logistic regressions (folds split by image) that add attention to position and token log-probability.

3.4 Image-attention boost (E4)

In each chosen layer, for every query q after the last image token and every key k, a'(q, k) = a(q, k)(1 + alpha 1[k in image]) / sum_k' a(q, k')(1 + alpha 1[k' in image]). The image tokens' own rows are untouched. This is equivalent to adding log(1 + alpha) to the image keys' pre-softmax logits, a constant shift, whereas PAI's alpha|logit| depends on each logit. We implement it by wrapping Transformers' eager attention function. Unit tests on a tiny random Qwen2-VL check that alpha = 0 gives bit-identical logits and generations, that rows still sum to 1, that the boosted share follows s' = s(1 + alpha)/(1 + alpha s), and that rows in a left-padded batch are boosted exactly as when run alone. We sweep alpha in {0.5, 1, 2, 4} and layer bands {0-7, 8-15, 16-27, 0-27} plus the band chosen by E3 on dev. The **selection rule was fixed before the sweep**: among settings whose dev recall drops by at most 2 points and dev POPE F1 by at most 1 point, take the lowest dev CHAIR-i; if none qualifies, take the lowest dev CHAIR-i overall and report that the criteria were not met.

PAI-style comparison (E4b). To compare with a published method, we also implement PAI's attention step [5]: before the softmax, each image key's logit z becomes z + alpha|z|, only in the row of the token being generated. PAI also contrasts the output with an image-free forward pass; we omit that step, so the comparison is between the two ways of moving attention onto the image. Its grid is alpha in {0.05, 0.1, 0.2, 0.5} on all layers except the first two (2-27) or on layers 6-13. Larger alpha is not useful on this model: at alpha = 0.5 on layers 2-27, dev captions shrink to 24 words, and in a smoke test they turned into lists of bounding-box coordinates. The setting is chosen on dev with the same rule as above, and each test set is then run once. Two unit tests check that only the last query row changes and that batched rows match rows run alone.

3.5 Comparison baseline and abstention (E5)

As a simple comparison we use a cautious POPE prompt (" Answer yes only if the object is clearly visible in the image, otherwise answer no."), fixed before any test run. For abstention, an extra forward pass records s_l at the last prompt token (where the yes/no decision is made) and the probabilities of "yes" and "no". We compare three scores for deciding which questions to answer: mean image attention over the 8-layer band that best separates right from wrong answers *on dev*, answer confidence |P(yes) - P(no)|, and a random order. We plot accuracy against coverage, report the area under that curve, and apply a threshold set on dev (keeping 80% of dev questions) unchanged to each test split. The same idea is applied to captions by dropping object mentions whose image attention falls below dev quantiles.

4. Experiments and results

All test numbers below come from one run of each frozen method; dev numbers were used only to make the choices described in Section 3. Appendix A lists the per-experiment commands and output files.

4.1 E0: sanity checks

In bfloat16, 8 of 8 checks pass: answers parse as yes/no, greedy decoding is deterministic, the image-token span matches the image grid, attention rows sum to one and no tensor contains NaN. In float16, 5 checks fail: the image-attention share is defined in layer 0 but NaN in every later layer, and every answer is "!!!!!" (Appendix B). An attention study run in float16 would have silently produced empty results, so we report all results in bfloat16.

4.2 E1-E2: how much does the unmodified model hallucinate?

Split	Method	Acc	Prec	Rec	F1	Yes %	McNemar p
random	Baseline	88.2	98.1	77.9	86.8	39.7	-
	Cautious prompt	88.8	97.4	79.8	87.7	41.0	0.009
	Attention boost	87.9	98.1	77.3	86.5	39.4	0.243
	PAI-style boost	88.1	98.1	77.7	86.7	39.6	0.688
popular	Baseline	87.2	95.7	77.9	85.9	40.7	-
	Cautious prompt	87.6	94.5	79.8	86.5	42.2	0.228
	Attention boost	87.1	96.1	77.3	85.7	40.2	0.644
	PAI-style boost	87.2	96.0	77.7	85.9	40.5	1.000
adversarial	Baseline	85.8	92.5	77.9	84.5	42.1	-
	Cautious prompt	86.1	91.4	79.8	85.2	43.7	0.242
	Attention boost	85.6	92.6	77.3	84.3	41.7	0.471
	PAI-style boost	85.7	92.7	77.6	84.5	41.9	1.000

Table 1. POPE test results (3,000 questions per split, %). Baseline = unmodified model; Cautious prompt = E5 comparison; Attention boost = alpha = 2, layers 6-13, selected on dev; PAI-style = alpha = 0.2, layers 6-13, selected on dev by the same rule. McNemar p: exact test against the baseline on the same questions.

Qwen2-VL-2B is accurate on POPE (88.2%, 87.2% and 85.8% on the random, popular and adversarial splits; Table 1). As the absent objects get harder, precision drops from 98.1% to 92.5%, while recall is 77.9% on every split because the present-object questions are shared across splits. Unlike the LLaVA-style models that POPE was designed around, which answer "yes" too often [1], this model is *conservative*: it says "yes" to only 39.7-42.1% of questions although half are positive. Of its 355 errors on the random split, 332 are misses ("no" to an object that is there) and only 23 are hallucinated "yes" answers (adversarial: 95 of 427). In free captions the picture is different. 36.2% of the 500 test captions contain at least one hallucinated object and 8.0% of object mentions are hallucinated (CHAIR-i), at 72.3% object recall (Table 2). Captions are long (173 words on average) and 35.6% reach the 256-token limit.

Method	CHAIR-s	CHAIR-i	Recall	Words	Objects/caption	Truncated %
Baseline	36.2	8.0	72.3	172.8	8.53	35.6
Attention boost	31.8	7.5	71.4	144.9	7.66	14.2
PAI-style boost	32.4	7.1	71.1	157.9	7.96	21.6
Boost - baseline (95% CI)	-4.4 [-8.4, -0.8]	-0.5 [-1.6, 0.5]	-0.9 [-2.5, 0.8]			
PAI-style - baseline (95% CI)	-3.8 [-7.8, 0.0]	-0.9 [-1.8, 0.1]	-1.2 [-2.8, 0.3]			
Matched length: baseline	31.0	7.2	69.1		7.37	
Matched length: boost	31.4	7.4	71.0		7.49	
Matched length: difference	0.4 [-3.6, 4.2]	0.2 [-0.8, 1.2]	1.9 [0.0, 3.9]			

Table 2. CHAIR on the 500 test images (%). Differences = method minus baseline with a paired image-level bootstrap 95% CI. Length-matched rows (attention boost; PAI-style values are in the text): for each image both captions are cut to the shorter caption's word count before scoring, which removes the effect of caption length.

4.3 E3: do hallucinated objects receive less image attention?

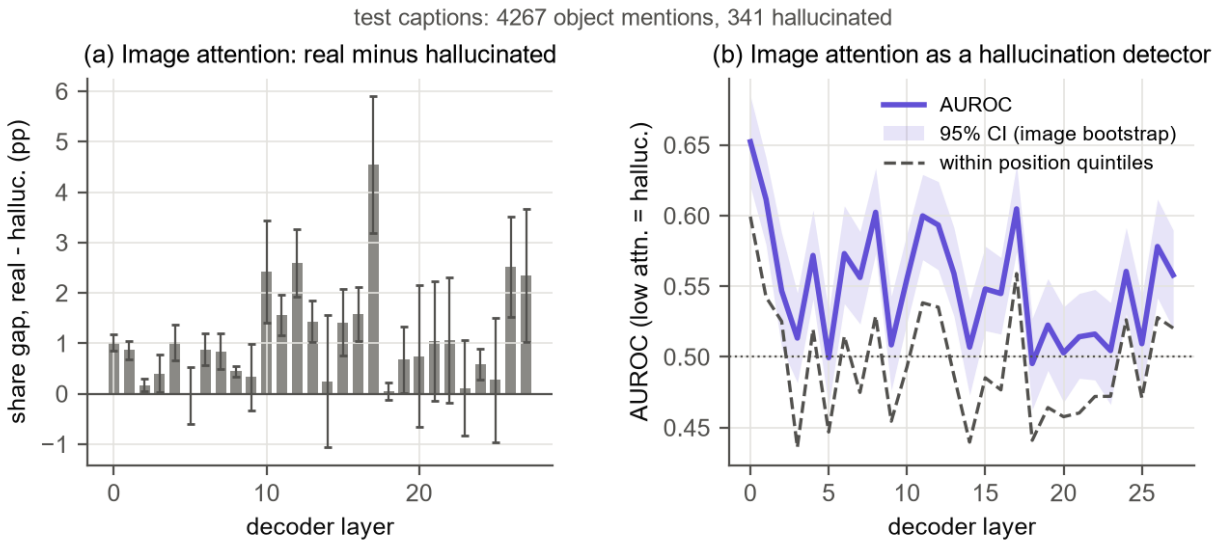


Figure 1. E3 on the test captions (4267 object mentions, 341 hallucinated). (a) Mean image-attention share at the position predicting the object's first token, real minus hallucinated (95% CI). (b) AUROC of low image attention as a detector of hallucinated mentions, with an image-bootstrap 95% CI and the AUROC within caption-position quintiles. The dev figure (Appendix C) looks the same.

Hallucinated mentions do receive slightly less image attention: the gap is positive in 27 of 28 layers on the test captions, and largest in layer 17 (27.1% vs 22.5% of attention on the image; Figure 1a). As a detector it is weak. The best single layer is layer 0 (AUROC 0.652, 95% CI 0.621-0.684), the same layer as on dev (0.643), and many layers after 13 are close to chance. Much of the signal is position: hallucinations come late in long captions, where every token attends less to the image. Within position quintiles the AUROC falls to 0.599 at layer 0 and at most 0.559 elsewhere. In cross-validated detectors (Table 3), the model's own token log-probability is the strongest single feature, and adding image attention (band 6-13, chosen on dev) to position and log-probability does not help (test AUROC 0.679 without, 0.677 with).

Features (logistic regression, 5-fold CV by image)	AUROC dev	AUROC test
Position in caption	0.604	0.623
Token log-probability	0.667	0.648
Image attention (band)	0.540	0.562
Position + log-prob	0.700	0.679
Position + log-prob + attention	0.697	0.677

Table 3. AUROC for detecting hallucinated object mentions (logistic regression, 5-fold cross-validation with folds split by image). Dev: 1624 mentions (134 hallucinated); test: 4267 (341).

4.4 E4: does boosting image attention reduce hallucination?

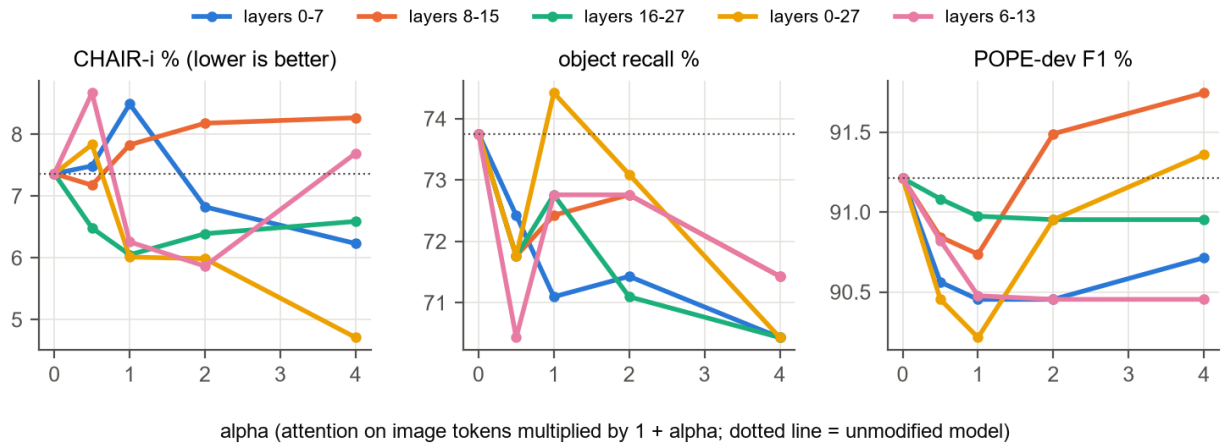


Figure 2. E4 sweep on dev (100 captions and the 902 dev POPE-style questions). Each line is a layer band; the dotted line is the unmodified model. The full table is in Appendix D.

On dev, only boosting all 28 layers lowers CHAIR-i fairly steadily as alpha grows, to 4.7% at alpha = 4 (baseline 7.4%; Figure 2). The other bands cross the baseline several times, which shows how noisy CHAIR-i is on 100 captions: one point of CHAIR-i is about 8 object mentions. What changes consistently is length. The all-layer boost at alpha = 4 cuts captions from 173 to 120 words and recall by 3.3 points, which breaks the recall constraint. POPE F1 stays within one point of the baseline everywhere. The fixed rule selected alpha = 2 on layers 6-13, the band E3 chose (dev CHAIR-i 7.4% to 5.9%, recall 73.8% to 72.8%). We later repeated the sweep on all 200 dev captions (Table D3). With the same rule it would choose alpha = 2 on layers 0-27 instead (dev CHAIR-i 8.3% to 6.5%, captions 173 to 127 words). Our setting still lowers dev CHAIR-i there (8.3% to 7.5%), but the choice of band is not stable. We did not run the new choice on the test sets, because this check was made after the test runs. A rerun on a second GPU (RTX 5060 Ti, same software versions) shows the same instability. POPE reproduces closely (99.4-99.8% identical answers, accuracy within 0.5 points), but captions differ word for word because bfloat16 rounding differs between GPU models, which alone moves CHAIR on 100 dev captions by up to about 2 points. On that GPU the same rule selects alpha = 4 on layers 16-27, and our frozen setting no longer lowers dev CHAIR-i. Only 6 of the 20 settings meet the selection rule on both GPUs; alpha = 2 on layers 0-27, the 200-caption pick, is one of them.

On the test set the boost changes POPE by at most 0.3 accuracy points, and no split differs significantly (McNemar p >= 0.24; Table 1). On CHAIR (Table 2), CHAIR-s changes by -4.4 points (95% CI -8.4 to -0.8), but CHAIR-i changes by only -0.5 points (95% CI -1.6 to +0.5) and recall by -0.9 points (95% CI -2.5 to +0.8). Captions become 28 words shorter, and only 14.2% (instead of 35.6%) reach the token limit. Shorter captions contain fewer objects, so fewer of them contain a hallucination even at the same hallucination rate per mention. The length-matched comparison confirms this. When both captions of an image are cut to the same length, the CHAIR-s difference is +0.4 points (95% CI -3.6 to +4.2) and the CHAIR-i difference is +0.2 points (95% CI -0.8 to +1.2), while recall is +1.9 points *higher* (95% CI +0.0 to +3.9). The boost makes the model describe the same objects in fewer words; it does not make it hallucinate less often (Figure 3).



Baseline (213 words, 2 hallucinated mentions): The image depicts a scene in what appears to be a zoo or wildlife sanctuary. The setting is an outdoor enclosure with a dirt ground and a wooden fence. There are several giraffes in the enclosure, standing and moving around. The giraffes are of different sizes, indicating they are of various ages. Some of them are closer to the camera, while others are further away, giving a sense of depth to the scene. In the background, there is a building with a brown roof and a wooden exterior. The building appears to be a shelter or a small structure, possibly for the giraffes to rest or shelter from the elements. The building is partially obscured ‖ by the trees and the giraffes, but it is visible enough to identify its purpose. There are also some trees and shrubs in the background, providing a naturalistic environment for the giraffes. The trees have tall, slender trunks and green foliage, which contrasts with the brown and beige tones of the building and the giraffes. The sky is visible in the upper part of the image, suggesting that the enclosure is located in a region with clear weather conditions. In the foreground, there is a small **bird** on the ground, possibly a ground-dwelling species. The **bird** is small

Attention boost (115 words, 0 hallucinated mentions): The image depicts a scene in a zoo or wildlife sanctuary, featuring a group of giraffes in an enclosure. The giraffes are standing on a dirt ground, with some of them reaching up towards the tall trees in the background. The enclosure is fenced with wooden posts and a metal gate, providing a barrier between the giraffes and the visitors. There are several giraffes visible in the image, with some standing closer to the foreground and others further back. The giraffes appear to be in a relaxed state, enjoying their time in the enclosure. The overall atmosphere of the image is calm and serene, capturing the natural beauty of these magnificent creatures in their habitat.

Figure 3. The length effect on one test image (COCO val2014 image 64313; annotated objects: giraffe, zebra). Orange bold: object CHAIR scores as hallucinated; blue: correct object; ‖ marks where the boosted caption ends (word 115). The baseline's hallucinated bird comes at word 202, in a caption cut off by the token limit; neither caption mentions the zebra. It is one of 15

test images where every baseline hallucination comes after the boosted caption has ended and the boosted caption has none (chosen by hand among them as a genuine hallucination).

PAI-style comparison (E4b). On dev, the PAI-style step behaves like our boost (Appendix D, Table D2). On layers 2-27 at alpha = 0.2 it lowers CHAIR-i from 7.4% to 4.7%, but captions shrink from 173 to 95 words and recall falls by 3.3 points, which breaks the recall constraint. The rule selected alpha = 0.2 on layers 6-13 (dev CHAIR-i 7.4% to 7.0%, recall 73.8% to 73.1%). On test it changes POPE accuracy by at most 0.1 points (McNemar p >= 0.69; Table 1). On CHAIR (Table 2), CHAIR-s changes by -3.8 points (95% CI -7.8 to +0.0), CHAIR-i by -0.9 points (95% CI -1.8 to +0.1) and recall by -1.2 points (95% CI -2.8 to +0.3), with captions 15 words shorter. After length matching, the differences shrink to -0.4 points (95% CI -4.2 to +3.4) for CHAIR-s and -0.4 points (95% CI -1.4 to +0.6) for CHAIR-i. So PAI's attention step, tuned under the same protocol, gives the same picture as our boost: POPE does not change, and the small CHAIR gains come largely from shorter captions.

4.5 E5: a prompt baseline and attention-based abstention

The cautious prompt raises POPE accuracy by +0.6, +0.4 and +0.3 points (significant only on random, p = 0.009). It works the opposite way to its wording: it *raises* the yes-ratio and recall (77.9% to 79.8%) and slightly lowers precision. For a conservative model, a prompt that mentions the object being "clearly visible" seems to help it accept objects it would otherwise miss.

Split	Acc (all)	Score used to rank questions	AUROC (wrong)	Area acc-cov	Coverage @ dev thr.	Acc @ dev thr.
random	88.2	low image attention, layers 20-27	0.579	91.0	80.0	89.1
		answer confidence	0.845	96.9	73.6	95.7
		high attention, layer 20 (exploratory)	0.766	94.9	-	-
		random order	0.500	88.2	-	-
popular	87.2	low image attention, layers 20-27	0.569	90.0	78.5	87.9
		answer confidence	0.817	96.0	68.3	95.2
		high attention, layer 20 (exploratory)	0.729	93.8	-	-
		random order	0.488	86.9	-	-
adversarial	85.8	low image attention, layers 20-27	0.566	88.4	78.8	86.4
		answer confidence	0.793	94.9	62.8	94.4
		high attention, layer 20 (exploratory)	0.693	92.2	-	-
		random order	0.502	85.9	-	-

Table 4. E5 abstention on POPE test. Questions are answered in order of the score; AUROC (wrong) measures how well the score ranks wrong answers last. The thresholds (80% coverage on dev) are fixed on dev. The exploratory row uses a direction found by inspecting dev and was not part of the planned comparison.

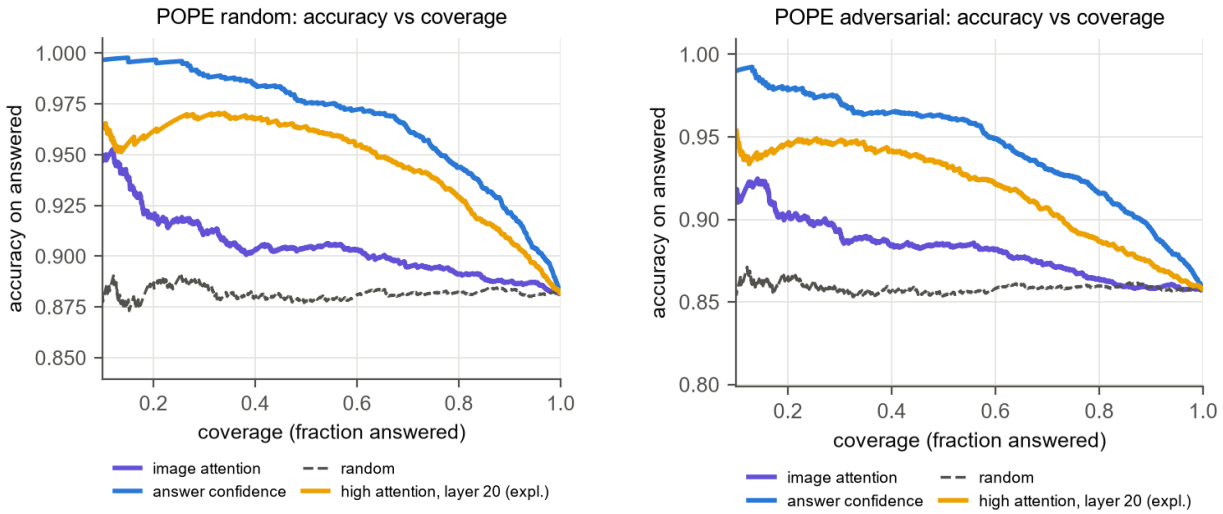


Figure 4. Accuracy of the answered questions against coverage on POPE random (left) and adversarial (right).

For abstention, answer confidence is far better than image attention (Table 4, Figure 4). Its AUROC for wrong answers is 0.793-0.845, against 0.566-0.579 for low attention in the band chosen on dev (layers 20-27). With the dev threshold, confidence answers 74% of random-split questions at 95.7% accuracy. Attention answers 80% at 89.1%, barely above the 88.2% obtained by answering everything. Looking at dev, the strongest

attention signal ran the *other* way: in layer 20, wrong answers had *more* image attention. This held on test (AUROC 0.766 random, 0.693 adversarial; 0.767 among "no" answers only). The model's typical error is a miss, made while it is attending to the image strongly but fails to find the object. Even this post-hoc signal is weaker than confidence. For captions, dropping the 20% of object mentions with the lowest image attention (dev quantile) lowers CHAIR-i only from 8.0% to 7.5%, at a recall cost of 1.5 points.

5. Discussion

5.1 What the results say about the hypotheses

- **H1 is only weakly supported.** Hallucinated objects receive slightly less image attention in most layers, but the effect is small (best AUROC 0.652), mostly explained by where the object appears in the caption, and adds nothing to the model's own token confidence. Image attention is a symptom of position more than a cause of hallucination.
- **H2 is not supported.** The selected boost leaves POPE unchanged and CHAIR-i within noise. Its significant CHAIR-s reduction disappears once caption length is controlled. The PAI-style step, tuned the same way, behaves the same (Section 4.4).
- **H3 is partly supported.** On dev the only clear CHAIR-i reduction (all layers, alpha = 4) came with much shorter captions and lower recall (Figure 2), and the setting allowed by the constraints did not lower CHAIR-i on test. Because the boost changed no POPE split, we cannot say whether the adversarial split benefits least; the cautious prompt did help the adversarial split least.

5.2 Failure cases (E6)



Figure 5. E6 cases from the test set: POPE answers the boost made wrong, POPE errors that persist, and objects the boost introduced into captions (heuristic cause in brackets).

We grouped errors with simple heuristics and then read the samples (Figure 5). Of the 341 POPE-random questions both methods get wrong, 254 ask about an object covering less than 1% of the image. In the samples we read, POPE errors are misses of small, dark or partly hidden objects (a chair in a dark room, a bowl on a crowded table). Some are annotation errors: one "orange" label refers to an orange cat. Among the 84 objects the boost introduced into captions, 44 frequently co-occur with objects that are present, most often a person or a table (a table under a camera tripod, a microwave and toaster in a kitchen). This is the text-prior pattern that motivated the study. Many others are not hallucinations at all, but CHAIR scores them as hallucinated: "alert drivers to slow down" (person) on a road marking, "a younger or smaller adult" (person) for an elephant, "a children's book" (person), and a person in a bear costume (bear). To size this problem we read the caption context of 30 randomly sampled CHAIR hallucinations from the baseline: **18 of 30** are scoring artefacts. These include word senses ("truck bed", "toilet bowl", "a remote landscape", "train cars"), loose synonyms ("container" as bowl, "streetlights" as traffic light), colours ("the vase is orange"), negations ("no visible people") and objects in pictures. The remaining 12 look like genuine hallucinations or missing annotations, often hedged ("possibly a phone").

5.3 What the negative results mean

Three things explain why moving attention onto the image, with our boost or with PAI's attention step, did not help this model, unlike the 7B LLaVA-style models in PAI [5]. (We omit PAI's image-free contrast, which may account for part of its reported gains.) First, Qwen2-VL-2B's failure mode on POPE is the opposite of the one these methods target. It misses objects far more often than it invents them, and its misses come with *high*, not low, image attention. Pushing more attention onto the image cannot fix a perception failure. Second, in captions the boost changes *style* (shorter, more list-like descriptions that reach the objects sooner) rather than grounding. The per-mention hallucination rate is unchanged, and CHAIR-s rewards shorter captions. This is why we report recall, length and a length-matched comparison next to CHAIR-s. Third, the measurement is noisy. CHAIR-i on 100 dev captions moves by more than a point between neighbouring settings, a sweep on 200 dev captions picks a different layer band, the same sweep on another GPU picks yet another setting, and if the hand check is representative, more than half of the hallucination labels are scoring artefacts. That noise weakens any detector, including E3's, and can hide small real effects.

5.4 Limitations

We study one 2B model on COCO images only, with greedy decoding and one caption per image. The sweep used 100 dev captions; with 200 captions, or on another GPU, the same rule picks a different setting (Section 4.4). A third of the baseline captions reached the 256-token limit. CHAIR depends on a fixed synonym list and misses or misreads objects (Section 5.2). Unlike the original NLTK tokenisation, our re-implementation also splits hyphenated words ("man-made" counts as man); this touches 20 of 4267 baseline mentions and moves CHAIR-i by 0.1 points for the baseline and 0.1 for the boost. Our hand check covered 30 mentions and used the caption text, not the image. Attention weights are not a full explanation of model behaviour. Our boost is one specific, constant rescaling. We re-implemented only PAI's attention step, without its image-free contrast, and not VCD or OPERA, so we cannot rank against the full published methods. The decision-time attention and the exploratory layer-20 result describe correlations, not causes.

6. Conclusion

We asked whether Qwen2-VL-2B pays less attention to the image when it hallucinates an object, and whether paying more attention helps. With a strict dev/test protocol the answer is: only slightly, and no. Hallucinated mentions receive a little less image attention (AUROC up to 0.652), mostly because they come late in captions. A training-free image-attention boost, and PAI's attention step tuned the same way, leave POPE and per-mention hallucination unchanged; their apparent CHAIR-s gains are caption-length effects (length-matched CHAIR-i +0.2 and -0.4 points). The model's own answer confidence is a much better abstention signal than image attention. For this small, conservative model, most POPE errors are misses rather than inventions, so methods built to stop language-prior hallucinations in larger models do not transfer automatically. Evaluations should report recall, caption length and scoring artefacts next to CHAIR. Promising next steps are a better hallucination scorer (for example an open-vocabulary detector or human labels) and checking whether the reversed attention signal for misses holds in other models.

AI Use Statement

We used Claude (Anthropic), through Claude Code, to help plan the project, write and debug the evaluation, CHAIR, attention-analysis, intervention, sweep, abstention and plotting code, run the experiment scripts on a team member's GPU, and draft and edit this report. We checked the code with unit tests (attention rows summing to one, alpha = 0 reproducing the original outputs bit for bit, batched rows matching single rows, CHAIR parsing and metrics on hand-made examples, AUROC against scikit-learn) and with the E0 sanity checks, which also caught the float16 overflow. We read captions with their hallucinated words marked and the failure cases by hand; Claude proposed the labels for the 30-mention CHAIR hand check (Section 5.2) from the caption text, and we checked each one. We checked every reference against the original paper. Every number in this report is produced by our scripts from saved JSON files and listed in RUN_LOG.md. Our team chose the research question, decided the experimental protocol and selection rules, reviewed the results and wrote the interpretation. [Team: edit this statement so that every sentence is true for what you did.]

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Appendix

A. Reproducing the results

The README gives full instructions. The main commands, run from the repository root with SC4001_ROOT pointing at a folder for data and results, are:

- `pip install -r requirements.txt` (Python 3.10+, PyTorch with CUDA; versions in README.md)
- `python scripts/prepare_data.py` (COCO val2014 images and annotations, POPE files, CHAIR synonyms, dev split)
- `bash scripts/run_all.sh` (E0 to E6 in order; resumable; about 8 hours on an RTX 4060 Laptop GPU)
- `python scripts/make_report.py` (every table and numbers.json from the saved outputs, no GPU)
- `node report/build_report.js --overwrite; powershell -File report/to_pdf.ps1` (first draft of this report, then edited in Word)
- Reproducibility check: parts of the pipeline were rerun from a fresh clone on an RTX 5060 Ti (data preparation, E0, POPE random, the dev sweep, 48 test captions and E3 attention on 40 test captions; outputs in results/reruns/rtx5060ti/). The data splits and POPE questions are identical, 99.4-99.8% of POPE answers match the original run, the E0 checks pass, and E3 attention shares agree to about 1% (relative); captions differ word for word because of bfloat16 rounding (Section 4.4). All tables and figures rebuild exactly from the saved outputs. A second rerun of POPE random on a V100 of the CCDS TC1 cluster, in float32 (the V100 has no bfloat16) and from the committed code, gives 88.4% accuracy (original 88.2%); 2,968 of 3,000 answers (98.9%) are identical, and the two runs do not differ significantly (McNemar p = 0.22).

Outputs: results/pope/<method>/<split>.jsonl (per question), results/chair/<method>/chair_test_scored.jsonl (per caption, with every object mention), results/attention/ and results/analysis/ (E3, E5, E6), results/sweep/ (E4), results/report/numbers.json (every number in this report) and RUN_LOG.md (what was run, when and on which hardware).

B. E0 sanity checks

Check	bfloat16	float16
answers look like yes/no	pass	FAIL
present object gets 'yes'	pass	FAIL
greedy decoding is deterministic	pass	pass
image-token positions	pass	pass
one attention map per layer	pass	pass
attention has no NaN/inf	pass	FAIL
attention rows sum to 1	pass	FAIL
logits have no NaN/inf (float16 overflow check)	pass	FAIL

Table B1. E0 checks on the RTX 4060 Laptop GPU (results/sanity/). In float16 every attention map after layer 0 is NaN, so the float16 checks that still pass say little. The saved float16 report marks "present object gets 'yes'" as pass because the original check accepted any answer parsed as "yes", including "!!!!!"; the corrected check (RUN_LOG, 2026-11-11) marks it FAIL.

C. E3 on the dev captions

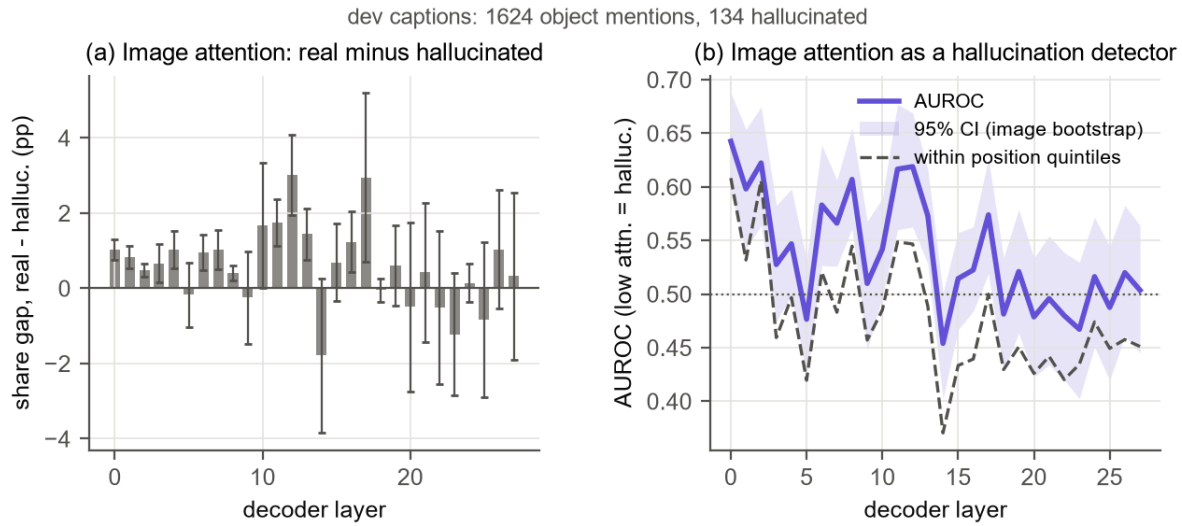


Figure C1. As Figure 1, on the dev captions (1624 mentions, 134 hallucinated). The layer band 6-13 used by E4 was chosen from this figure's data.

D. E4 sweep tables (dev)

alpha	Layers	POPE acc	POPE F1	Yes %	CHAIR-s	CHAIR-i	Recall	Words
0 (base)	none	91.8	91.2	43.3	31.0	7.4	73.8	173
0.5	0-7	91.2	90.6	42.8	31.0	7.5	72.4	167
1	0-7	91.1	90.5	42.9	35.0	8.5	71.1	158
2	0-7	91.1	90.5	42.9	31.0	6.8	71.4	144
4	0-7	91.4	90.7	43.1	28.0	6.2	70.4	128
0.5	8-15	91.5	90.8	43.2	34.0	7.2	71.8	174
1	8-15	91.4	90.7	43.3	33.0	7.8	72.4	172
2	8-15	92.0	91.5	43.8	38.0	8.2	72.8	167
4	8-15	92.2	91.7	44.0	35.0	8.3	71.4	149
0.5	16-27	91.7	91.1	43.2	32.0	6.5	71.8	176
1	16-27	91.6	91.0	43.3	27.0	6.0	72.8	178
2	16-27	91.6	91.0	43.1	29.0	6.4	71.1	180
4	16-27	91.6	91.0	43.1	27.0	6.6	70.4	176
0.5	0-27	91.1	90.5	42.9	35.0	7.8	71.8	165
1	0-27	90.9	90.2	42.9	32.0	6.0	74.4	152
2	0-27	91.6	91.0	43.1	28.0	6.0	73.1	132
4	0-27	91.9	91.4	43.7	24.0	4.7	70.4	120
0.5	6-13	91.5	90.8	43.0	38.0	8.7	70.4	171
1	6-13	91.1	90.5	43.1	27.0	6.3	72.8	164
2	6-13 *	91.1	90.5	42.9	28.0	5.9	72.8	150
4	6-13	91.1	90.5	42.9	29.0	7.7	71.4	128

Table D1. All sweep settings on dev (%), 100 captions. * = selected by the fixed rule.

alpha	Layers	POPE acc	POPE F1	Yes %	CHAIR-s	CHAIR-i	Recall	Words
0 (base)	none	91.8	91.2	43.3	31.0	7.4	73.8	173
0.05	2-27	91.5	90.8	43.0	33.0	7.4	71.8	171
0.1	2-27	91.4	90.7	42.7	27.0	6.1	70.1	146
0.2	2-27	91.5	90.8	42.6	23.0	4.7	70.4	95
0.5	2-27	89.9	89.2	43.0	9.0	5.0	47.5	24
0.05	6-13	91.8	91.2	43.3	34.0	8.1	73.8	174
0.1	6-13	91.7	91.1	43.2	36.0	7.7	72.4	173
0.2	6-13 *	91.8	91.2	43.3	31.0	7.0	73.1	169
0.5	6-13	91.8	91.2	43.6	22.0	7.0	51.8	75

Table D2. PAI-style sweep on dev (%), 100 captions. * = selected by the same rule.

alpha	Layers	POPE acc	POPE F1	Yes %	CHAIR-s	CHAIR-i	Recall	Words
0 (base)	none	91.8	91.2	43.3	32.5	8.3	69.9	173

alpha	Layers	POPE acc	POPE F1	Yes %	CHAIR-s	CHAIR-i	Recall	Words
0.5	0-7	91.2	90.6	42.8	32.5	7.9	71.6	167
1	0-7	91.1	90.5	42.9	32.0	7.9	70.3	154
2	0-7	91.1	90.5	42.9	30.0	7.4	69.1	138
4	0-7	91.4	90.7	43.1	29.5	7.4	68.9	124
0.5	8-15	91.5	90.8	43.2	36.5	7.6	70.5	170
1	8-15	91.4	90.7	43.3	34.5	8.1	71.0	168
2	8-15	92.0	91.5	43.8	34.5	8.5	70.3	161
4	8-15	92.2	91.7	44.0	33.0	7.8	71.5	142
0.5	16-27	91.7	91.1	43.2	35.5	8.0	72.0	174
1	16-27	91.6	91.0	43.3	30.5	6.7	72.7	177
2	16-27	91.6	91.0	43.1	34.5	7.6	70.8	177
4	16-27	91.6	91.0	43.1	32.0	7.2	70.1	176
0.5	0-27	91.1	90.5	42.9	35.5	7.7	71.5	161
1	0-27	90.9	90.2	42.9	32.5	6.6	72.0	147
2	0-27 *	91.6	91.0	43.1	29.5	6.5	70.6	127
4	0-27	91.9	91.4	43.7	26.5	6.7	66.9	107
0.5	6-13	91.5	90.8	43.0	38.0	9.3	72.0	170
1	6-13	91.1	90.5	43.1	31.5	7.5	71.0	164
2	6-13	91.1	90.5	42.9	31.5	7.5	70.6	148
4	6-13	91.1	90.5	42.9	33.5	8.6	69.8	123

Table D3. Robustness check: the E4 sweep on all 200 dev captions (%). * = what the rule would select; this run was made after the test runs and was not used to choose anything.

E. Abstention on the other splits

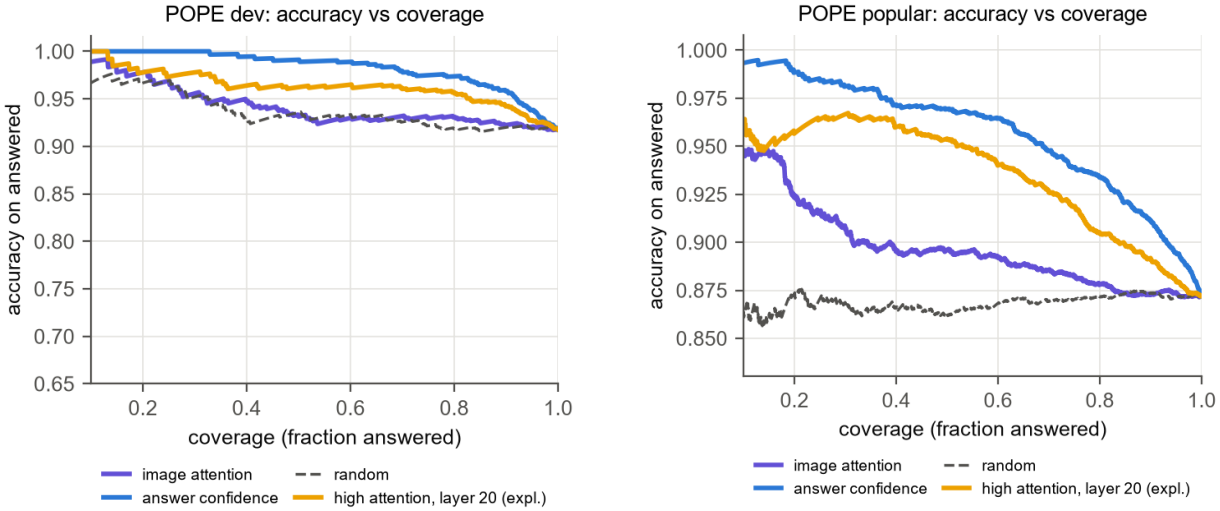


Figure E1. Accuracy against coverage on the dev questions (left; where the band and thresholds were chosen) and POPE popular (right).